

> Boston Dynamics pulled half a mile of wire from its truck-unloading robot to cut downtime

Stretch Gen 2 was redesigned from 100,000+ hours of field data. What else changed, and what is still unproven?

> 01 — WHAT IT DOES

Boston Dynamics' Stretch empties truck trailers box by box



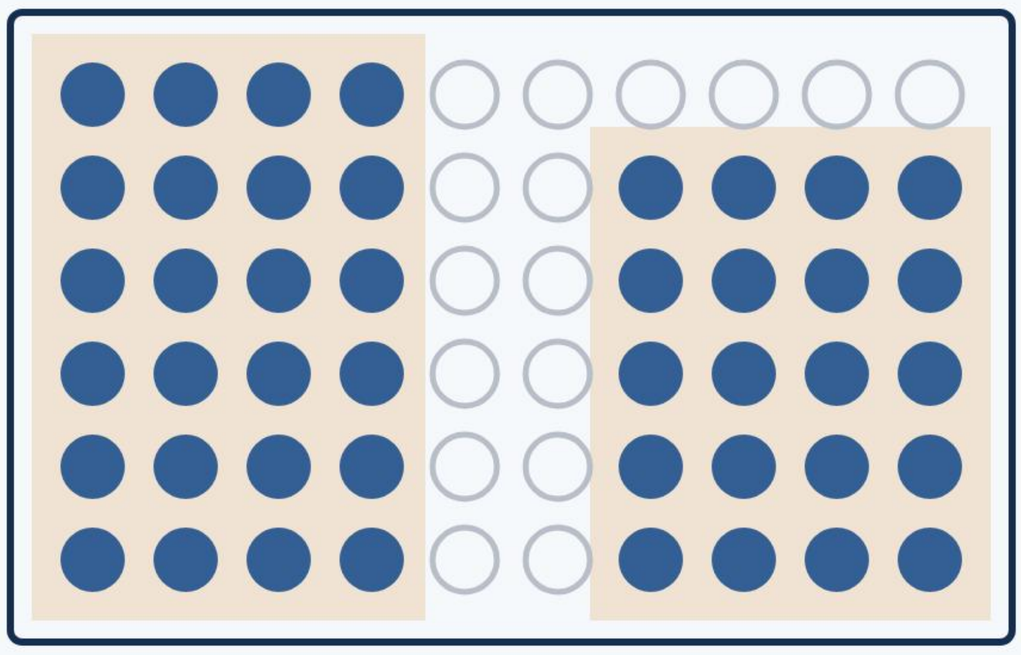
Stretch is a mobile manipulator, a robot arm on a wheeled base. It drives into a trailer and moves boxes onto a conveyor with a suction gripper. Above, the first generation.

Boston Dynamics calls it one of the worst warehouse jobs: hot, cold, heavy. Gen 2 already runs at several customer sites.

> 02 — THE GRIPPER_

Stretch Gen 2 gives every suction cup its own valve

Arm motion takes up most of each pick, so grabbing several boxes per swing raises throughput. Each cup senses contact and only pulls vacuum on a box. That saves pump power and lets Stretch drop the boxes onto the conveyor one by one.



Touching a box:
valve open, suction

Touching air:
valve shut, no vacuum wasted

Illustration: gripper from below

> 03 — WHAT BROKE_

Stretch Gen 2 cuts 81% of its internal cabling

Cables flex with every arm motion until they wear out. Boston Dynamics calls them the biggest source of intermittent electrical faults.

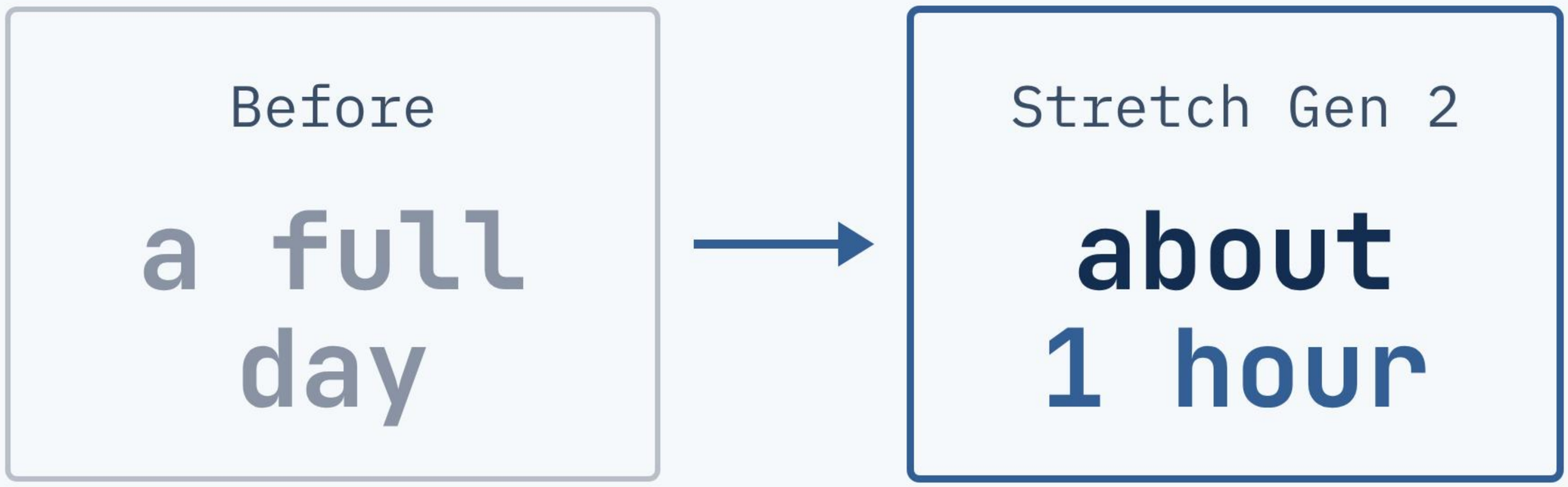
The cut removes about half a mile of wire, nearly nine football fields end to end. Hyundai sourced the new cable harness through its car supply chain, and the connectors come from cars too: they latch, so vibration can't shake them apart.

- Automotive-grade connectors that latch shut
- Joints that spin past 360°, no hard stops
- Emergency stop set lower, out of boxes' way

> 04 — SERVICE_

A drive-module swap on Stretch Gen 2 takes about an hour

Swapping one full swerve drive



A swerve drive is a wheel module that both drives and steers. Standard high-strength fasteners replace complex harnesses, and onboard diagnostics let a customer's own maintenance staff handle routine fixes.

> 05 — WHAT TO WATCH_

Stretch Gen 2's 1,000 cases an hour is a peak rate

That's one box every 3.6 seconds, and it depends on box weight and how neatly the trailer is stacked. Software ships about monthly; the latest update raised unload rates 24% across the fielded fleet, Boston Dynamics says.

The launch named no price, unit count or customer.

"Every hour a robot sits idle is an hour of work that piles up on the dock."

— BRIAN DONNELLY, BOSTON DYNAMICS, NOV 3, 2026

> END OF TRANSMISSION_



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